

```

82 • select * from incentives;
83 • show tables;_
84 • SELECT DISTINCT A.EMPNO
85 FROM ASSIGNED_TO A
86 JOIN PROJECT P ON A.PNO = P.PNO
87 ✖ WHERE P.PLOC IN ('Bengaluru', 'Hyderabad', 'Mysuru');++

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Tables_in_employee			
assigned_to			
dept			
employee			
incentives			
project			

```

84 • SELECT DISTINCT A.EMPNO
85 FROM ASSIGNED_TO A
86 JOIN PROJECT P ON A.PNO = P.PNO
87 WHERE P.PLOC IN ('Bengaluru', 'Hyderabad', 'Mysuru');
88 • select e.empno from employee e where empno not in(select
89
90 • SELECT E.ENAME, E.EMPNO, E.DEPTNO, E.JOB_ROLE, D.LOCAT
91 FROM EMPLOYEE E

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
EMPNO			
101			
102			
103			

```

Result 2 ✖
87 WHERE P.PLOC IN ( Bengaluru , Hyderabad , Mysuru );
88 • select e.empno from employee e where empno not in(select i.empno from incentives i);
89
90 • SELECT E.ENAME, E.EMPNO, E.DEPTNO, E.JOB_ROLE, D.LOCATION AS DLOC, P.PLOC
91 FROM EMPLOYEE E

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
empno			
101			

```

98 #WEEK03
99 • Select e.empno, e.ename, d.dname, d.dloc
100 from employee e
101 join dept d on e.deptno = d.deptno;
102
103 • Select e.empno, e.ename from employee e
104 where e.empno not in (select e.empno from assigned_to);
105
106 • SELECT p.pno, p.pname, COUNT(a.empno) AS headcount
107 FROM project p
108 LEFT JOIN assigned_to a ON p.pno = a.pno
109 GROUP BY p.pno, p.pname;

```

Result Grid |  Filter Rows: | Export:  Wrap Cell Content: 

	empno	ename	dname	dloc
▶	101	Alice	HR	Bengaluru
	102	Bob	Finance	Hyderabad
	103	Charlie	Engineering	Mysuru
	104	David	Marketing	Chennai
	105	Eva	Sales	Delhi

```

3 • Select e.empno, e.ename from employee e
4 where e.empno not in (select e.empno from assigned_to);
5
6 • SELECT p.pno, p.pname, COUNT(a.empno) AS headcount
7 FROM project p
8 LEFT JOIN assigned_to a ON p.pno = a.pno
9 GROUP BY p.pno, p.pname;
0
1
2 • select e.deptno,max(e.sal) as maximum salary,avg(e.sal) a

```

Result Grid |  Filter Rows: | Export:  Wrap Cell Content: 

empno	ename
-------	-------

```

105
106 • SELECT p.pno, p.pname, COUNT(a.empno) AS headcount
107 FROM project p
108 LEFT JOIN assigned_to a ON p.pno = a.pno
109 GROUP BY p.pno, p.pname;
110
111
112 • select e.deptno,max(e.sal) as maximum_salary,avg(e.sal) as a

```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 

	pno	pname	headcount
▶	201	Apollo	1
	202	Zeus	1
	203	Hermes	1
	204	Athena	1
	205	Poseidon	1

Result 6 x

```

.1
.2 • select e.deptno,max(e.sal) as maximum_salary,avg(e.sal) as average_salary from employee e group by(e.deptno);
.3
.4 • select i.empno,sum(i.incentive_amount) as total_incentive from incentives i group by (i.empno);
.5
.6 • SELECT e.empno, e.ename, e.deptno
.7 FROM employee e
.8 JOIN assigned_to a ON e.empno = a.empno

```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 

deptno	maximum_salary	average_salary
10	60000.00	60000.000000
20	55000.00	55000.000000
30	58000.00	58000.000000
40	62000.00	62000.000000
50	50000.00	50000.000000

```

113
114 • select i.empno,sum(i.incentive_amount) as total_incentive from incentives i group by (i.empno);
115
116 • SELECT e.empno, e.ename, e.deptno
117 FROM employee e
118 JOIN assigned_to a ON e.empno = a.empno

```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 

empno	total_incentive
▶ 102	5000.00
	103 4000.00
	104 2500.00
	105 3500.00

105
106 • SEL
107 FRQ
108 LEF
109 GRG
110
111
112 • sel
Result Grid 
pno pname
▶ 201 Apollo
202 Zeus

```

115
116 • SELECT e.empno, e.ename, e.deptno
117 FROM employee e
118 JOIN assigned_to a ON e.empno = a.empno
119 JOIN project p ON a.pno = p.pno
120 WHERE p.pname = 'Apollo';
121

```

Result Grid			Filter Rows:	Export: 	Wrap Cell Content: 
	empno	ename	deptno		
▶	101	Alice	10		

```

1
2 • SELECT d.deptno, d.dname
3 FROM dept d
4 WHERE d.deptno NOT IN (SELECT e.deptno FROM employee e);_
5
6 • SELECT m.empno, m.ename, COUNT(e.empno) AS report_count
7 FROM employee m
8 JOIN employee e ON m.empno = e.mgr_no
9 GROUP BY m.empno, m.ename;

```

Result Grid			Filter Rows:	Export: 	Wrap Cell Content: 
	deptno	dname			

- ```

SELECT m.empno, m.ename, COUNT(e.empno) AS report_count
FROM employee m
JOIN employee e ON m.empno = e.mgr_no
GROUP BY m.empno, m.ename;

```

Alt Grid | Filter Rows: | Export: | Wrap Cell Content: |

| empno | ename   | report_count |
|-------|---------|--------------|
| 101   | Alice   | 2            |
| 102   | Bob     | 1            |
| 103   | Charlie | 1            |

- ```

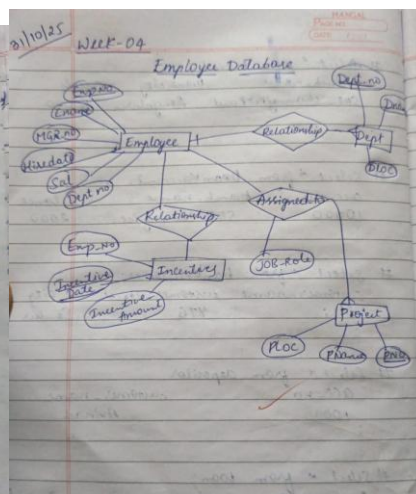
SELECT a.empno, COUNT(a.pno) AS projectcount
FROM assigned_to a
GROUP BY a.empno
HAVING COUNT(a.pno) > 1;

```

Alt Grid | Filter Rows: | Export: | Wrap Cell Content: |

empno	projectcount
-------	--------------

INDEX				
M.S. SWATHI DBMS LAB BOOK				
3 Sem. GSE Roll No. 1010101010				
BMSCE				
Sr No.	Date	Title	Page No.	Teacher Sign
1	20/10/25	WEEK-01 INSURANCE DB		
2	20/10/25	WEEK-02 INSURANCE		
3	20/10/25	WEEK-03 Bank database		
4	20/10/25	WEEK-04 Employee database		
5	20/10/25	WEEK-05 Employee database (not given)		



Create database employee;

use employee;

create table employee (emp-no int primary key, ename varchar(50), deptno int, mgr-no int, hiredate date, sal decimal(10,2), foreign key (dept-no) references (dept-no));

Create table project (pro int primary key, pname varchar(50), ploc varchar(50));

Create table assigned-to (emp-no int, pro int, job-role varchar(50), primary key (emp-no, pro), foreign key (emp-no) references employee (emp-no), foreign key (pro) references project (pro));

create table incentives (emp-no int, incentive-date date, incentive-amount decimal(10,2), primary key (emp-no, incentive-date), foreign key (emp-no) references employee (emp-no));

create table dept (dept-no int primary key, d-name varchar(50), d-loc varchar(50));

insert into dept values (10, 'HR', 'Bengaluru'), (20, 'Finance', 'Hyderabad'), (30, 'Engineering', 'Mysuru'), (40, 'Marketing', 'Chennai'), (50, 'Sales', 'Delhi');

insert into employee values (101, 'Alice', 10, NULL, '2018-06-01', 60000), (102, 'Bob', 20, 101, '2019-07-15', 55000), (103, 'Charlie', 30, 101, '2020-03-20', 58000), (104, 'David', 40, 102, '2021-01-10', 62000), (105, 'Eva', 50, 103, '2022-09-05', 50000);

insert into project values (201, 'Apollo', 'Bengaluru'), (202, 'Zeus', 'Hyderabad'), (203, 'Hermes', 'Mysuru'), (204, 'Athena', 'Chennai'), (205, 'Poseidon', 'Delhi');

insert into assigned-to values (101, 201, 'Manager'), (102, 202, 'Analyst'), (103, 203, 'Developer'), (104, 204, 'Marketer'), (105, 205, 'Sales Executive'), (103, 102, 'HR');

insert into incentives values (102, '2019-01-10', 3000), (103, '2019-01-15', 4000), (104, '2019-01-20', 2500), (105, '2019-01-25', 3500), (102, '2020-02-10', 2000);

Select * from dept;

dept-no	d-name	d-loc
10	HR	Bengaluru
20	Finance	Hyderabad
30	Engineering	Mysuru
40	Marketing	Chennai
50	Sales	Delhi
Null	Null	Null

Select * from employee;

emp-no	ename	deptno	mgr-no	hiredate	sal
101	Alice	10	Null	2018-06-01	60000.00
102	Bob	20	101	2019-07-15	55000.00
103	Charlie	30	101	2020-03-20	58000.00
104	David	40	102	2021-01-10	62000.00
105	Eva	50	103	2022-09-05	50000.00

Select * from project;

pro	pname	ploc
201	Apollo	Bengaluru
202	Zeus	Hyderabad
203	Hermes	Mysuru
204	Athena	Chennai
205	Poseidon	Delhi

Select * from assigned-to;

emp-no	pro	job-role
101	201	Manager
102	202	Analyst
103	203	Developer
104	204	Marketer
105	205	Sales Executive

Select * from incentives;

emp-no	incentive-date	incentive-amount
102	2019-01-10	3000.00
103	2020-02-10	2000.00
103	2019-01-15	4000.00
104	2019-01-20	2500.00
105	2019-01-25	3500.00

Select distinct a.emp-no from assigned-to a join project p on a.pro = p.pro where p.ploc in ('Bengaluru', 'Hyderabad', 'Mysuru');

emp-no
101
102
103

Get employee Id's of those employees who did not receive incentives.
 # Select empno from employee e where empno not in (select empno from incentives i);
 q/p
 empno
 102
 Q) Write a SQL query to find the employee name, number, dept, jobrole, department location and project location who are working for a project location same as his/her department location.
 # select e.empno, e.ename, d.deptname, d.ploc as dept_location, a.jobrole, p.ploc as project_location from employee e, dept d, assigned-to a, project p, where e.dept-no = d.dept-no and e.empno = a.empno and a.pno = p.pno and d.ploc = p.ploc;

emp-no	ename	dname	dept location	Job role
101	Alice	IT	Bengaluru	Analyst
102	Bob	HR	Mysuru	Analyst
103	Charlie	Finance	Hyderabad	Developer
105	Eva	Admin	Chennai	Sales Executive

project_location
Bengaluru
Mysuru
Hyderabad
Chennai

WEEK 05:

Employee database MANGAL
PAGE NO. _____
DATE / /

#11/25 Week-05 more queries.

Q) List employees with their department name & location

select e.empno, e.ename, d.dname, d.loc from employee e join dept d on e.deptno = d.deptno;

Q/P

empno	ename	dname	loc
101	Alice	HR	Bangalore
102	Bob	Finance	Hyderabad
103	Charlie	Engineering	Kyurum
104	David	Marketing	Chennai
105	Eva	Sales	Delhi

Q) Employees who are not assigned to any project

select e.empno, e.ename from employee e where e.empno not in (select e.empno from assigned-to);

Q/P

emp-no	ename

Project database MANGAL
PAGE NO. _____
DATE / /

Project-wise

select p.pno, p.pname, count(a.empno) as headcount from project p left join assigned-to a on p.pno = a.pno group by p.pno = p.pname;

Q/P

pno	pname	headcount
201	Apollo	1
202	Zeus	1
203	Hermes	1
204	Athena	1
205	Poseidon	1

Q) Department wise average and max salary

select e.deptno, max(e.sal) as maximum salary, avg(e.sal) as average salary from employee e group by (e.deptno);

Q/P

deptno	maximum salary	average salary
10	60000.00	60000.000000
20	55000.00	55000.000000
30	58000.00	58000.000000
40	62000.00	62000.000000
50	50000.00	50000.000000

Incentives database MANGAL
PAGE NO. _____
DATE / /

Q) Total incentives earned by each employee

select i.empno, sum(i.incentive-amount) as total-incentive from incentives i group by (i.empno);

Q/P

emp-no	total-incentive
102	5000.00
103	4000.00
104	2500.00
105	3500.00

Q) Employees working on a given project

select e.empno, e.ename from employee e, project p, assigned-to a where e.empno = a.empno and e.pno = p.pno and p.pname = 'Apollo';

Q/P

empno	ename	deptno
101	Alice	10

Managers database MANGAL
PAGE NO. _____
DATE / /

Q) Departments with no employees

select d.deptno from deptno where d.deptno not in (select distinct e.deptno from employee e);

Q/P

deptno	dname

Q) Managers and their direct report counts

select m.mgrno, count(m.mgrno) as report-count from employee e, employee m where e.empno = m.mgrno group by (m.mgrno);

Q/P

empno	mgrno	report-count
101		2
102		1
103		1

8) Employees assigned to more than one project
 # select a.emp.no, e.name, count(*)
 as number-of-projects from employee e,
 assigned-to a where e.emp.no =
 a.emp.no group by (a.emp.no) having
 count(*) > 1;

(OR)
 select a.emp.no, count(a.p.no) as
 project-count from assigned-to a
 group by a.emp.no having count
 (a.p.no) > 1;

emp.no	project-count / number-of-projects
103	2

20
 2/11